University of Nottingham
UK | CHINA | MALAYSIA

School of Computer Science

A Level 1 Module Autumn 2024/2025

Mathematics for Computer Scientists

Instructions:

Time allowed: Two Hours

Answer ALL FOUR Questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so**.

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

No calculators are permitted in this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: NONE

Additional material: NONE

Information for Invigilators: NONE

Exam Papers must not be removed from the examination venue.

1. This question is about *Logic, Proofs*.

[**overall 25 marks**]

(a) What is a proposition? What is a tautology? What is a contradiction? (3 marks)

(b) Write down the truth tables for $(p \to q) \wedge (q \to r) \wedge (r \to p)$ and $p \wedge q \wedge r$. Use these tables to find a truth assignment for p, q, r so that $(p \to q) \wedge (q \to r) \wedge (r \to p)$ is true and $p \wedge q \wedge r$ is false. (5 marks)

(c) In the domain of polygons, let the predicate $P(x, y)$ be the statement "x has more sides than y", $Q(x)$ be "x is a square" and $R(x)$ be "x is a hexagon".

(i) Translate these two sentences into predicate logic: (4 marks)

(A) No square is a hexagon.

(B) All hexagons have more sides than any square.

(ii) Translate this statement into English: (4 marks)

(C) $\exists x(Q(x) \wedge \forall y(P(y, x) \to \neg Q(y)))$

(iii) Can statement (A) be inferred from statements (B) and (C)? If yes, then give a prove. If no, give a counter-example. (4 marks)

(d) Use mathematical induction to show that 5 divides $6^n - 1$ where n is a positive integer. (5 marks)

2. This question is about *Sets, Functions, Relations, Probability*.

[**overall 25 marks**]

(a) Let X and Y be the following sets:

$X = \{5, 8, 12, 15\}$
$Y = \{1, 8, 12, 13, 18\}$

State the results for the following operations:

(i) $|Y| - |X|$ (1 marks)
(ii) $X \cup Y$ (1 marks)
(iii) $Y - X$ (1 marks)
(iv) $\mathcal{P}(X \cap Y)$ (2 marks)

(b) Let $f(x)$ be a function from $\mathbb{R}$ to $\mathbb{R}$.
Is the function $f(x) = (x-1)(x-2)$ invertible? If yes, what is its inverse? (2 marks)

(c) Let $X = \{1, 2, 3\}$. Define a relation R on $X \times X$ as follows:

$$\forall a, b, c, d \quad (a, b)R(c, d) \quad iff \quad ad = bc$$

Prove that R is an equivalence relation on $X \times X$. (6 marks)

(d) Suppose there are 7 digits: 0,1,2,3,4,5,6. We use these 7 digits to form an integer with no repeated digits. Consider the following questions below.
(Hint: an integer should start with a non-zero digit)

(i) How many ways are there to form a 7-digit odd integer?
(2 marks)

(ii) How many ways are there to form a 4-digit integer that is divisible by 25?
(3 marks)

(e) Consider the following game. A person rolls a fair dice. This person receives a payment of $2n$ pounds where n is the number of the points obtained. For example, if the person has got 5 points on the dice, he/she receives 10 pounds cash.

i. The person needs to pay 9 pounds to roll the dice once.
What is the probability that the player will win more then he/she paid to play?
(3 marks)

ii. The person needs to pay 11 pounds to roll the dice twice. In this case, she/he wins according to the dice showing the highest points. For example, if the person rolls 3 and 1, then 3 is the highest roll and so the person wins $2 \times 3 = 6$ pounds.
What is the probability that the player will win more than she/he paid to play?
(4 marks)

3. This question is about *systems of linear equations and vector spaces*.

[**overall 25 marks**]

(a) Consider the following system of linear equations:

$$\begin{aligned} 3x_1 + 2x_2 + x_4 &= 7 \\ 2x_2 - x_3 + x_4 &= 1 \\ x_1 - x_2 + 3x_3 &= 1 \\ 4x_2 - 2x_3 + 2x_4 &= 2 \end{aligned}$$

(i) Write the complete matrix $\mathbf{A}^c$ which represents this system of linear equations. (3 marks)

(ii) Compute the rank of the complete matrix $\mathbf{A}^c$. Show full working. (3 marks)

(iii) Use Rouché-Capelli Theorem to determine if this system of linear equations is compatible, determined, undetermined or incompatible. (2 marks)

(iv) Suppose that $x_3 = 0$. Then use Cramer's Method to compute the solution for x_2. (5 marks)

(b) Consider the following sets:

$$\begin{aligned} E &= \{(x, y, z) \in \mathbb{R}^3 \mid 2x - y = z\} \\ F &= \{(x, y, z) \in \mathbb{R}^3 \mid 2x \geq y + z\} \\ G &= \{(x, y, 0) \in \mathbb{R}^3 \mid 2x = y\} \\ H &= \{(x, y) \in \mathbb{R}^2 \mid 2x = y\} \end{aligned}$$

Let + and . be the usual arithmetic addition and product operators, respectively.

(i) Show whether or not E and F are closed with respect to internal "+" and external "." composition laws. (4 marks)

(ii) Which of these statements are true and which are false? (8 marks)

S1. $(E, +, .)$ is a vector space.
S2. $(F, +, .)$ is a vector space.
S3. $(G, +, .)$ is a vector space.
S4. $(H, +, .)$ is a vector space.
S5. $E \subset F$
S6. $(E, +, .)$ is a vector subspace of $(F, +, .)$.
S7. $(G, +, .)$ is a vector subspace of $(E, +, .)$.
S8. $(H, +, .)$ is a vector subspace of $(E, +, .)$.

4. This question is about *linear mappings*.

[**overall 25 marks**]

Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear mapping $f(x, y, z) = (3z, 2y + z, x - y - z)$.
Suppose there exists a linear mapping $g : \mathbb{R}^3 \to \mathbb{R}^3$ such that for all $(x, y, z) \in \mathbb{R}^3$,

$$g(f(x, y, z)) = (x, y, z).$$

(a) Compute $f(3, 2, 1)$ (1 mark)

(b) Is f an endopmorphism? Explain why it is or isn't. (2 marks)

(c) Show that f is bijective. (5 marks)

(d) Compute ker(f). (2 marks)

(e) Compute dim(ker(f)). (1 mark)

(f) Use the Rank-Nullity Theorem to compute dim(Im(f)). (2 marks)

(g) Let $U \subset \mathbb{R}^3$. Show that the image of U through g is the inverse image of U through f. That is,

$$g(U) = f^{-1}(U).$$

(5 marks)

(h) Write the matrix representation **A** of linear mapping f. (2 marks)

(i) Compute the matrix representation **B** of linear mapping g. (3 marks)

(j) What is the matrix product **BA**? (2 marks)